Vibrational Deduction Transformer (VDT)

A Variational Autoencoder with Graph Wiring
Latent Space and Associative Spectral Memory

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Abstract

We introduce the *Vibrational Deduction Transformer* (VDT), a generative architecture that treats the learning process as a vibrational system grounded in Rayleigh's *Theory of Sound*. The VDT is a deductive transformer that leverages feature-space graph Laplacian: the learning process is guided by the Laplacian eigenbasis as harmonics of a spring/string vibrating system endowed with a characteristic density matrix. The decoder maps a stochastic latent code $\mathbf{z}$ to a *spectral loading matrix* $\mathbf{W} = \mathbf{U}_{1:q}\,\mathrm{diag}(\boldsymbol{\omega})\,\mathbf{S}$, where $\mathbf{U}_{1:q}$ are the q lowest-frequency eigenvectors of a frozen ArrowSpace graph Laplacian index $\mathbf{L}(\mathcal{I})$, and $\mathbf{S}$ is a learnable coefficient matrix in that eigenbasis. An associative Hopfield-type memory is appended to the spectral artefact emitted after training, wiring the vibrational representation into a downstream transformer. The objective is a four-term Evidence Lower Bound (ELBO) comprising a reconstruction term, an isotropic latent KL, an eigenvalue-weighted spectral-basis KL (spectral Occam's razor), and a Gamma-vs-Exponential mode-frequency KL for spectral band selection. Benchmarks on CORA, PubMed, and synthetic spring-network graphs demonstrate entropy-controlled spectral graph generation and strong linear-probe accuracy of frozen latent representations.

Contents

1 Introduction

Transformer-based generative models typically organise learned representations in a flat isotropic Gaussian latent space, discarding the spectral structure of the data. With ArrowSpace [**?**] we demonstrated that any vector space can span a semantic web of relations from its feature-space via graph reconstruction algorithm (Graph Wiring [5]); this structure carries critical semantic information that a flat prior cannot capture.

The Vibrational Deduction Transformer (VDT) draws direct inspiration from the *Lattice Deduction Transformer* (LDT) of Davis et al. [3]. The LDT is a recurrent transformer that approximates logically sound deduction by projecting its latent state through a lattice between forward passes. Grounded in abstract interpretation [4], it models information states as elements of a lattice with a top element $\top$ representing no information and a bottom element $\perp$ representing inconsistency; each forward pass of the recurrent transformer constitutes a sound deduction step that descends from $\top$ toward more informative lattice states. The model is trained on-policy via a Solve loop—the same stochastic search used at inference—and supervised by a domain-agnostic alpha operator α that aggregates the valid solutions still consistent with the current state. An 800K-parameter LDT achieves 100% accuracy on Sudoku-Extreme benchmarks while frontier large language models score 0%, demonstrating that structured iterative refinement over a discrete lattice can vastly outperform chain-of-thought paradigms on hard logical reasoning tasks. The VDT adapts this deductive, iterative-refinement philosophy to the continuous spectral domain: rather than refining a discrete candidate lattice, VDT refines a vibrational latent representation guided by the graph Laplacian eigenbasis.

Based on this, the Vibrational Deduction Transformer (VDT) is motivated by three converging lines of inquiry:

1. **Rayleigh's vibrational analogy.** Rayleigh [1] showed that the natural oscillation modes of a continuous medium are the eigensolutions of its governing differential operator. Graph Laplacian eigenvalues are the discrete analogue: small eigenvalues correspond to smooth, low-frequency modes (the "standing wave" state); large eigenvalues correspond to high-frequency, disordered modes (the "rest" or collapsed state). The VDT treats the *low-frequency oscillation state as the low-entropy prior* and high-frequency disorder as high entropy, directly reversing the usual thermodynamic framing: the natural state of the string is to oscillate, not to be still.

2. **Real-valued quantum amplitudes.** Aaronson [2] observes in Chapter 9 of *Quantum Computing Since Democritus* that the essential features of quantum probability—interference, entanglement structure, the Born rule—are recoverable from signed real amplitudes without complex numbers, provided the update rule is unitary. The VDT implements this insight: the entries of the loading matrix $\mathbf{W}$ can be positive or negative, modulating the relative magnitude of Laplacian eigenvector contributions and producing constructive or destructive interference in the heat kernel through the eigenvector structure of $\mathbf{L}(\mathbf{z})$.

3. **ArrowSpace / Graph Wiring.** The ArrowSpace computational framework [**?**] defines a differentiable Laplacian builder and a spectral cost (J_{freq}) that penalises high-frequency wiring. The VDT replaces this hard penalty with a *variational* counterpart—a KL divergence over spectral loadings and mode weights—yielding a proper Bayesian ELBO.

The paper follows the conceptual flow applied in [8].

The paper is organised as follows. Section 2 reviews prerequisite material. Section 3 presents the full VDT architecture. Section 4 derives the four-term ELBO. Section 5 describes the spectral artefact and associative memory. Section 6 covers the stability analysis. Section 7 reports empirical results. Section 8 discusses related work. Section 9 concludes.

2 Background

2.1 Graph Laplacians and Spectral Graph Theory

Given an undirected weighted graph $G = (V, E, w)$ with $N = |V|$ nodes, the *combinatorial Laplacian* is $\mathbf{L} = \mathbf{D} - \mathbf{A}$, where $\mathbf{D}$ is the degree matrix and $\mathbf{A}$ the weighted adjacency. The normalised Laplacian $\tilde{\mathbf{L}} = \mathbf{D}^{-1/2}\mathbf{L}\mathbf{D}^{-1/2}$ has eigenvalues $0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_N \leq 2$. The eigenvectors $\{\mathbf{u}_k\}$ form an orthonormal basis (the *graph Fourier basis*), and the *Rayleigh quotient*

$$\mathcal{R}(\mathbf{v}) = \frac{\mathbf{v}^\intercal \mathbf{L}\mathbf{v}}{\mathbf{v}^\intercal \mathbf{v}} \tag{1}$$

gives the frequency (smoothness cost) of signal $\mathbf{v}$ on the graph [9]. This is the discrete analogue of Rayleigh's classical frequency functional for continuous media [1].

2.2 Real Quantum Amplitudes and Signed Interference

Aaronson [2] finds plausible that the observable consequences of quantum mechanics—interference, entanglement structure, Born-rule probabilities—can be derived from the assumption of signed real probability amplitudes, provided the time-evolution operator is a *real orthogonal* (rather than complex unitary) matrix. The key insight is:

> "If we allow *negative* probability amplitudes, then we get quantum-like interference without complex numbers." [2]

The VDT operationalises this at the level of the *loading matrix*: entries of $\mathbf{W} \in \mathbb{R}^{N \times q}$ can be positive or negative, modulating the relative magnitude and sign of each eigenvector's contribution to the row embeddings of $\mathbf{L}(\mathbf{z})$. Edge weights themselves remain strictly positive (Section 3.4); the signed character of $\mathbf{W}$ instead reshapes the eigenvector decomposition of $\mathbf{L}(\mathbf{z})$, inducing constructive or destructive interference in the heat kernel $\mathbf{K}_\tau = \mathbf{U}\,\exp(-t\boldsymbol{\Lambda})\,\mathbf{U}^\intercal$ through the geometry of the learned eigenvectors.

3 Architecture

3.1 Overview

The VDT is a deductive transformer that leverages feature-space graph Laplacian: its generative path is mediated by a learned graph wiring constrained to the eigenbasis of an ArrowSpace index $\mathcal{I}$. Figure 1 summarises the data flow.

3.2 WiringEncoder

The encoder is a multi-layer perceptron (MLP) augmented with a λ-*fingerprint*—a fixed scalar summary of the Laplacian spectrum that encodes the global spectral geometry of the graph:

$$\text{fingerprint}(\mathbf{L}(\mathcal{I})) \;=\; f(\lambda_1, \ldots, \lambda_q) \in \mathbb{R}^{d_{\text{fp}}}, \tag{2}$$

where f is a deterministic aggregation function (e.g. normalised eigenvalue histogram). The encoder produces:

$$\mathbf{z},\, \boldsymbol{\mu},\, \log \boldsymbol{\sigma}^2,\, \log \mathbf{a},\, \log \mathbf{b} \;=\; \text{WiringEncoder}(\mathbf{x},\, \mathbf{L}(\mathcal{I})), \tag{3}$$

where $\mathbf{z} \in \mathbb{R}^{d_z}$ is the reparameterised VAE latent, $(\mathbf{a}, \mathbf{b}) \in \mathbb{R}^q_{>0} \times \mathbb{R}^q_{>0}$ are the shape and rate parameters of the Gamma variational posterior over mode weights.

3.3 SpectralLoadingDecoder

A linear projection $\mathbf{z}_q = \mathbf{W}_{z\to q}\,\mathbf{z} \in \mathbb{R}^q$ bridges the VAE latent dimension d_z to the spectral mode dimension q. The spectral loading decoder then produces:

$$\mathbf{S} = \text{S\_net}(\mathbf{z}_q) \in \mathbb{R}^{q\times q}, \quad \text{posterior mean of spectral coefficients,} \tag{4}$$

$$\log \boldsymbol{\sigma}^2_{\mathbf{S}} = \text{log\_var\_S\_head}(\mathbf{z}_q) \in \mathbb{R}^{q\times q}, \quad \textit{independent} \text{ posterior log-variance,} \tag{5}$$

$$\boldsymbol{\omega} = \exp(\text{omega\_net}(\mathbf{z}_q)) \in \mathbb{R}^q_{>0}, \quad \text{mode weights,} \tag{6}$$

$$\mathbf{W} = \mathbf{U}_{1:q}\,\text{diag}(\boldsymbol{\omega})\,\mathbf{S} \;\in\; \mathbb{R}^{N\times q}. \tag{7}$$

Remark 1. *In Eq.* (7)*, N denotes the* graph node count*—the row dimension of the eigenvector matrix $\mathbf{U}_{1:q} \in \mathbb{R}^{N\times q}$—and is* distinct *from the input feature dimension D used in Section 2.1. For the PPCA loading matrix $\mathbf{W}_{PPCA} \in \mathbb{R}^{D\times q}$ the row index ranges over feature coordinates; here it ranges over graph nodes. In CORA, for example,* $N = 2708$ *(nodes) while* $D = 1433$ *(bag-of-words feature dimension). The spectral loading matrix therefore has shape* $(2708 \times q)$*, projecting from the q-dimensional eigenbasis onto all N graph nodes—not onto feature coordinates.*

Equation (7) is the central structural commitment of the VDT: the loading matrix lives in the Laplacian eigenbasis. High-frequency directions are suppressed both by the mode weight prior (Section 4.4) and by the eigenvalue-weighted prior on $\mathbf{S}$ (Section 4.3). The independence of $\log \boldsymbol{\sigma}^2_{\mathbf{S}}$ from $\mathbf{S}$ is a deliberate design decision (fix #52): the old proxy $\log(\mathbf{S}^2 + \epsilon)$ conflated posterior mean and variance, producing biased KL gradients.

3.4 DifferentiableLaplacian

The loading matrix $\mathbf{W} \in \mathbb{R}^{N\times q}$ is converted to a graph Laplacian by:

$$w_{ij}(\mathbf{W}) = w_{ij}^{\text{base}} \cdot \sigma\Big(-\|\mathbf{W}_i - \mathbf{W}_j\|^2\Big), \quad \mathbf{L}(\mathbf{z}) = \mathbf{D}(\mathbf{W}) - \mathbf{A}(\mathbf{W}), \tag{8}$$

where σ is the logistic sigmoid, w_{ij}^{base} is a frozen base-graph weight, and $\mathbf{W}_i \in \mathbb{R}^q$ denotes the i-th row of $\mathbf{W}$ (the spectral embedding of node i). Since $\|\mathbf{W}_i - \mathbf{W}_j\|^2 \geq 0$ by definition, $\sigma(-\|\mathbf{W}_i - \mathbf{W}_j\|^2) \in (0, 0.5]$, so all edge weights are *strictly positive* and lie in $(0,\, w_{ij}^{\text{base}}/2]$. The construction remains fully differentiable through $\mathbf{W}$.

Quantum-like interference in the heat kernel does not arise here from negative edge weights. Instead, it arises from the *signed structure of* $\mathbf{W}$: because the entries of $\mathbf{W} = \mathbf{U}_{1:q}\,\text{diag}(\boldsymbol{\omega})\,\mathbf{S}$ can be positive or negative, rows $\mathbf{W}_i$ and $\mathbf{W}_j$ may be positively or negatively correlated in the eigenvector subspace. These signed correlations reshape the eigenvector decomposition of $\mathbf{L}(\mathbf{z})$: modes whose eigenvectors align with anti-correlated row pairs carry larger Laplacian eigenvalues and are exponentially suppressed in the heat kernel $\mathbf{K}_\tau = \mathbf{U}\,\exp(-t\boldsymbol{\Lambda})\,\mathbf{U}^\intercal$, producing constructive or destructive-like interference in the diffusion pathways between nodes—the real-amplitude counterpart of complex phase cancellation [2].

3.5 DiffusionDecoder (TauModeDiffusion)

The decoder applies a truncated heat kernel to the embedding table $\mathbf{E}$ and returns the heat-diffused row for each queried node:

$$\mathbf{K}_\tau = \mathbf{U}_\tau\,\exp(-\tau\,\boldsymbol{\Lambda}_\tau)\,\mathbf{U}_\tau^\intercal, \qquad \hat{\mathbf{x}} = \big(\mathbf{K}_\tau\,\mathbf{E}\big)_{[\text{node\_idx}]}, \tag{9}$$

where the summation is truncated to the leading τ_{modes} eigenpairs of $\mathbf{L}(\mathbf{z})$, and $\mathbf{E} \in \mathbb{R}^{N\times D}$ is the frozen node embedding table. The reconstruction is purely spectral: no MLP residual correction is applied inside `TauModeDiffusion` or in `WiringAutoencoder.forward()` after the diffusion step. The heat kernel $\mathbf{K}_\tau$ is a real symmetric positive-semi-definite operator—the real-valued analogue of a quantum channel [2, 13, 20].

4 The Four-Term ELBO

The VDT objective is the negative ELBO:

$$\boxed{\mathcal{L}_{\text{VDT}} = \underbrace{-\mathbb{E}[\log p(\mathbf{x}|\mathbf{z},\mathbf{W})]}_{\text{NLL recon}} + \underbrace{D_{\text{KL}}(q(\mathbf{z}) \,\|\, \mathcal{N}(\mathbf{0},\mathbf{I}))}_{\text{latent KL}} + \underbrace{D_{\text{KL}}(q(\mathbf{S}) \,\|\, p(\mathbf{S}|\mathcal{I}))}_{\text{spectral basis KL}} + \underbrace{D_{\text{KL}}(q(\boldsymbol{\omega}) \,\|\, p(\boldsymbol{\omega}|\tau,\boldsymbol{\Lambda}))}_{\text{taumode KL}}} \tag{10}$$

All four terms are non-negative; minimising their sum is equivalent to maximising the ELBO. We derive each term below.

4.1 Term 1: Reconstruction

Under the Gaussian output model the reconstruction term is:

$$-\mathbb{E}[\log p(\mathbf{x}|\mathbf{z},\mathbf{W})] = \frac{\|\mathbf{x}-\hat{\mathbf{x}}\|^2}{2\sigma^2} + \frac{D}{2}\log(2\pi\sigma^2). \tag{11}$$

In the implementation the constant $\frac{D}{2}\log(2\pi)$ is dropped; σ is a learnable scalar log-variance.

4.2 Term 2: Isotropic Latent KL

The standard VAE isotropic KL [11]:

$$D_{\text{KL}}(q(\mathbf{z}|\mathbf{x}) \,\|\, \mathcal{N}(\mathbf{0},\mathbf{I})) = -\frac{1}{2}\sum_{j=1}^{d_z}\left(1+\log\sigma_j^2-\mu_j^2-\sigma_j^2\right). \tag{12}$$

4.3 Term 3: Spectral-Basis KL

The coefficient matrix $\mathbf{S} \in \mathbb{R}^{q\times q}$ receives a *row-independent* Gaussian prior in which each entry S_{kj} is penalised by the k-th eigenvalue of the frozen Laplacian index:

$$p(S_{kj} \mid \mathcal{I}) \;=\; \mathcal{N}\!\left(0,\; \frac{1}{\lambda_S\,\lambda_k}\right) \qquad \forall\, k \in [q],\; j \in [q]. \tag{13}$$

Because the prior factorises over both indices, the joint log-density can be written as the *trace form*:

$$\log p(\mathbf{S} \mid \mathcal{I}) \;\propto\; -\frac{\lambda_S}{2}\,\mathrm{tr}(\mathbf{S}^\intercal\boldsymbol{\Lambda}_{1:q}\mathbf{S}) \;=\; -\frac{\lambda_S}{2}\sum_{k=1}^{q}\lambda_k\,\|\mathbf{S}_k\|^2, \tag{14}$$

where $\boldsymbol{\Lambda}_{1:q} = \mathrm{diag}(\lambda_1,\ldots,\lambda_q)$ and $\mathbf{S}_k \in \mathbb{R}^q$ is the k-th row of $\mathbf{S}$. Large eigenvalues penalise the corresponding rows of $\mathbf{S}$ more strongly—a *spectral Occam's razor*: high-frequency components of the loading matrix are shrunk towards zero.

The variational posterior is $q(\mathbf{S}) = \mathcal{N}(\mathbf{S}_0,\, \mathrm{diag}(\exp(\log\boldsymbol{\sigma}_{\mathbf{S}}^2)))$, where $\mathbf{S}_0$ and $\log\boldsymbol{\sigma}_{\mathbf{S}}^2$ are *independent* outputs of `SpectralLoadingDecoder` (fix #52). Under Gaussian–Gaussian conjugacy the KL is closed-form.

Remark 2. *The prior in Eq.* (13) *is* row-independent with identity column precision *($\boldsymbol{\Lambda}_{col} = \mathbf{I}$). A full* matrix-normal *prior would have the form* $\log p(\mathbf{S}) \propto -\frac{1}{2}\,\mathrm{tr}(\mathbf{S}^\intercal\boldsymbol{\Lambda}_{row}\mathbf{S}\boldsymbol{\Lambda}_{col})$ *with two separate precision matrices. When* $\boldsymbol{\Lambda}_{col} = \mathbf{I}$ *this reduces exactly to Eq.* (14)*. The implementation in* `vdt/spectral.py::spectral_basis_kl` *broadcasts the precision vector* `prec = lambda_S * eigvals_q` *over the column axis of* $\mathbf{S}$*, which is the direct computational realisation of Eq.* (13)*: one scalar precision per row* k*, shared across all* q *columns* j*.*

4.4 Term 4: taumode Frequency KL

Mode weights $\omega_k > 0$ receive an exponential prior indexed by frequency:

$$p(\omega_k|\tau, \lambda_k) = \text{Exponential}(\tau\lambda_k), \tag{15}$$

giving heavy support on low-frequency modes (small λ_k). The variational posterior is a Gamma: $q(\omega_k) = \text{Gamma}(a_k, b_k)$. The KL between a Gamma posterior and an Exponential prior is closed-form. Setting $r = \tau\lambda_k$ (prior rate), the derivation proceeds as:

$$\begin{aligned} \text{KL}\big(\text{Gamma}(a,b) \,\|\, \text{Exp}(r)\big) &= \mathbb{E}_q[\log q(\omega) - \log p(\omega)] \\ &= -H[\text{Gamma}(a,b)] - \log r + r\,\mathbb{E}_q[\omega] \end{aligned}$$

where the Gamma entropy is $H[\text{Gamma}(a,b)] = a - \log b + \ln\Gamma(a) + (1-a)\psi(a)$ and $\mathbb{E}_q[\omega] = a/b$. Substituting and simplifying gives:

$$D_{\text{KL}}(\text{Gamma}(a,b) \,\|\, \text{Exp}(r)) = \log b - \log r - \ln\Gamma(a) - (1-a)\psi(a) - a + \frac{a\,r}{b}, \tag{16}$$

where ψ is the digamma function and $r = \tau\lambda_k$. No Monte Carlo sampling is required. The implementation in `vdt/spectral.py::tau_mode_kl` follows Eq. (16) exactly; note the *negative* sign on $\ln\Gamma(a)$ and the additional $-a$ term, both of which arise from expanding the Gamma entropy.

Remark 3. *A common sign error writes* $+\ln\Gamma(a)$ *instead of* $-\ln\Gamma(a)$*. The correct sign follows immediately from the entropy expansion:* $H = a - \log b + \ln\Gamma(a) + (1-a)\psi(a)$*, which enters the KL with a* negative *prefactor (*$\text{KL} = -H + \ldots$*), flipping all signs, including the sign on* $\ln\Gamma(a)$*.*

4.5 Comparison with the Base Wiring AE

Table 1: ELBO structure: Base Wiring AE vs. VDT.

Term	**Wiring AE (base)**	**VDT**
Reconstruction	$-\log p(\mathbf{x}\|\mathbf{z})$	Same
Latent KL	$D_{\text{KL}}(q(\mathbf{z}) \,\|\, \mathcal{N}(\mathbf{0}, \mathbf{I}))$ isotropic	Same
Spectral basis	None	$D_{\text{KL}}(q(\mathbf{S}) \,\|\, p(\mathbf{S}\|\mathcal{I}))$ eigenvalue-weighted
Frequency	αJ_{freq} hard penalty	$D_{\text{KL}}(q(\boldsymbol{\omega}) \,\|\, \text{Exp}(\tau\lambda_k))$ variational

5 Spectral Artefact and Associative Memory

After training, the VDT operates in a two-phase regime that separates *spectral distillation* from *associative deployment*. The first phase extracts a compact, graph-indexed summary of everything the model has learned about the vibrational structure of the data; the second phase injects that summary as a structured prior into the weight matrices of a downstream transformer, enabling fast pattern-completion without retraining. Together, the two phases convert the continuous variational posterior into a discrete, inspectable memory store whose keys are guaranteed orthogonal by construction—a property inherited directly from the Laplacian eigenbasis.

5.1 Phase 1: Spectral Artefact Extraction

At the prior mean ($\mathbf{z} = \mathbf{0}$) the decoder emits a *spectral artefact*:

$$\mathcal{A}(\mathcal{I}) = \left\{\hat{\mathbf{W}},\ \hat{\boldsymbol{\omega}},\ \mathbf{S}_{\text{mem}}\right\}, \quad \mathbf{S}_{\text{mem}} = \sum_{k=1}^{q} \hat{\omega}_k \, d_\theta(\hat{\mathbf{w}}_k)\, \hat{\mathbf{w}}_k^\intercal, \tag{17}$$

where $\hat{\mathbf{w}}_k \in \mathbb{R}^N$ is the k-th column of $\hat{\mathbf{W}} \in \mathbb{R}^{N\times q}$ and $d_\theta(\hat{\mathbf{w}}_k)$ is the decoder response at spectral direction k. $\mathbf{S}_{\text{mem}}$ is a Hopfield outer-product memory matrix [19].

The artefact $\mathcal{A}(\mathcal{I})$ can be understood as a three-component *spectral certificate* of the index $\mathcal{I}$:

- $\hat{\mathbf{W}} \in \mathbb{R}^{N\times q}$: the MAP spectral loading matrix, whose k-th column is the learned spectral embedding of all N graph nodes along the k-th vibrational mode. Nodes that are spectrally similar—in the sense of the frozen Laplacian—cluster in the columns of $\hat{\mathbf{W}}$, encoding the community structure of the graph as a continuous vibrational pattern.

- $\hat{\boldsymbol{\omega}} \in \mathbb{R}^q_{>0}$: the posterior mode-weight means, which record which vibrational harmonics the model selected as informative. Low-frequency modes (small λ_k) receive higher weight under the taumode prior; the learned $\hat{\omega}_k$ profile therefore constitutes a *spectral fingerprint* that distinguishes graphs with different connectivity regimes.

- $\mathbf{S}_{\text{mem}} \in \mathbb{R}^{N\times N}$: the Hopfield memory matrix, constructed as a weighted sum of outer products $\hat{\mathbf{w}}_k\hat{\mathbf{w}}_k^\intercal$ scaled by both the mode weight $\hat{\omega}_k$ and the decoder response $d_\theta(\hat{\mathbf{w}}_k)$. This matrix encodes pairwise node associations in the spectral domain: entry (i, j) measures the degree to which nodes i and j co-activate across the retained vibrational modes, weighted by how strongly each mode is represented in the reconstruction.

Critically, the artefact is computed in a single deterministic forward pass at $\mathbf{z} = \mathbf{0}$, requiring no sampling and no additional training. It is a *frozen summary* of the posterior that can be serialised, transferred, or reused across different downstream tasks.

5.2 Phase 2: Spectral Memory Transformer

The transformer feed-forward and cross-attention value matrices are initialised from $\mathbf{S}_{\text{mem}}$, providing *long-term spectral prior memory.* Online delta-rule updates $\mathbf{S}_{\text{mem}} \leftarrow \mathbf{S}_{\text{mem}} + \eta\,(\mathbf{v} - \mathbf{S}_{\text{mem}}\mathbf{k})\mathbf{k}^\intercal$ write new associations without corrupting spectral key orthogonality. Key orthogonality (inherited from Laplacian eigenvectors) maximises retrieval SNR by construction.

The role of the associative memory is best understood as an *inline spectral prior*: rather than treating each transformer forward pass as starting from a blank context, the model begins every query with the graph's vibrational structure already imprinted in its weight matrices. This is the continuous-spectral counterpart of the LDT's lattice projection [3]: where the LDT constrains each recurrent step to remain within a lattice of logically consistent states, the VDT constrains the transformer's value space to remain aligned with the spectral geometry of the index $\mathcal{I}$.

The delta-rule update has two important properties. First, because the keys $\mathbf{k}$ are drawn from the Laplacian eigenbasis, they are mutually orthogonal: $\mathbf{k}_k^\intercal\mathbf{k}_{k'} = \delta_{kk'}$. Orthogonal keys eliminate cross-talk between stored patterns, so the retrieval signal-to-noise ratio scales as $\Theta(1)$ rather than degrading as $O(1/\sqrt{N_{\text{stored}}})$ under a random-key assumption [19]. Second, the update is *local* in the spectral domain: writing a new association $(\mathbf{v}, \mathbf{k})$ only modifies the projection of $\mathbf{S}_{\text{mem}}$ onto the subspace spanned by $\mathbf{k}$, leaving all orthogonal components—and hence all previously stored spectral patterns—intact. This gives the memory a *non-interfering write* property that classical full-matrix attention updates do not enjoy.

In practice, the initialisation of transformer value matrices from $\mathbf{S}_{\text{mem}}$ acts as a *warm start*: the model does not need to rediscover the spectral geometry of the graph from context tokens; instead, it inherits that geometry as a structural bias that can be refined but not easily overwritten.

This is particularly valuable in low-data regimes or when the downstream task involves the same graph index $\mathcal{I}$ used during VDT training, as the spectral prior concentrates probability mass on graph-consistent patterns before any task-specific supervision is applied.

5.3 Entropy-Controlled Generation

The spectral entropy of the generated Laplacian [21]:

$$H(\mathbf{L}(\mathbf{z})) = -\sum_{k=1}^{N} \bar{\lambda}_k \log \bar{\lambda}_k, \quad \bar{\lambda}_k = \frac{\lambda_k}{\sum_j \lambda_j}, \tag{18}$$

can be steered by conditioning $\mathbf{z}$ on a target entropy level, enabling *entropy-controlled spectral graph generation*—a capability absent in flat-prior VAEs.

The spectral entropy $H(\mathbf{L}(\mathbf{z}))$ serves as a coarse-grained *order parameter* for the generated graph: a low-entropy Laplacian corresponds to a graph whose eigenvalue mass is concentrated in a few low-frequency modes, indicating a highly structured, community-like connectivity; a high-entropy Laplacian corresponds to a near-uniform eigenspectrum, indicating a more random or expander-like wiring. Because the latent code $\mathbf{z}$ parametrises the loading matrix $\mathbf{W}$ and hence the full eigenspectrum of $\mathbf{L}(\mathbf{z})$, the encoder learns a direct mapping from input graphs to entropy levels during training, and at generation time the decoder can be queried at any $\mathbf{z}$ whose expected entropy—computed from the variational posterior—matches a desired target.

The associative memory $\mathbf{S}_{\text{mem}}$ interacts with this entropy-control mechanism in a complementary way. Each memory matrix is extracted at the prior mean $\mathbf{z} = \mathbf{0}$, which corresponds to the *average* spectral configuration of the training distribution. When a downstream transformer is initialised from $\mathbf{S}_{\text{mem}}$ and then presented with a query latent $\mathbf{z}^*$ associated with a particular entropy level, the delta-rule updates shift the memory toward the spectral patterns characteristic of that entropy regime without erasing the baseline structure. This allows the same memory module to serve as a *spectral prior library*: a single trained VDT can support multiple downstream tasks operating at different entropy levels, each refining the shared prior rather than training from scratch.

6 Stability Analysis

6.1 Six-Level Hierarchy

Training stability is analysed through a six-level hierarchy [7]; failures at lower levels must be resolved before diagnosing higher ones:

1. **Graph geometry.** $\lambda_{\max}(\mathbf{L}_f)$ finite; CFL condition satisfied; graph connectivity.
2. **Wave update per mode.** Damping coefficients $\gamma_k > 0$; underdamped/overdamped classification.
3. **Recurrent VDT dynamics.** Energy $\mathcal{E}_t$ monotone; spectral entropy H_{spectral} stable.
4. **Density matrix.** $\boldsymbol{\rho}_t^+$ positive semi-definite; $\|\boldsymbol{\rho}_t\|_F$ bounded.
5. **Optimiser.** Preconditioned gradient descent convergence: $\kappa(\mathbf{P}_{\sigma,M}) \ll \kappa(\mathbf{L}_f)$; step size $\eta < 2/L_{\sigma,M}$.
6. **Mode weights.** Active mode count $N_{\text{active}} = |\{k : \mathbb{E}[\omega_k] > \delta\}| \geq q_{\min}$; $\|\hat{\mathbf{W}}\|_F$ non-degenerate.

6.2 Mode Weight Collapse

For high-frequency modes with large $\tau\lambda_k$, the Exponential prior pushes $q(\omega_k)$ towards a point mass near zero. This is *correct behaviour* (mode selection), but if it extends to low-frequency modes the loading matrix degenerates. Two mitigations are applied:

- **Shape-parameter floor**: $a_k \geq a_{\min} = 0.1$ prevents full collapse.
- **Active-mode penalty**: a soft Lagrange term $\nu \cdot \mathrm{relu}(q_{\min} - N_{\mathrm{active}})$ maintains a minimum number of active modes.

6.3 Spectral Eigenvalue Floor

When the Fiedler eigenvalue $\lambda_2 \approx 0$ (nearly disconnected graph) the spectral-basis KL prior becomes nearly flat and produces high gradient variance. A floor applied in the KL computation only ($\lambda_k \leftarrow \max(\lambda_k, 10^{-3})$) prevents numerical instability without affecting the spectral geometry.

7 Experiments

7.1 Datasets

We evaluate on three benchmarks:

- **CORA** [15]: 2708 nodes, 5429 edges, 7 classes, 1433-dim bag-of-words features.
- **PubMed** [15]: 19717 nodes, 44338 edges, 3 classes.
- **Synthetic spring networks**: 400 randomly generated spring graphs with controllable spectral entropy.
- **Open Graph Benchmark (OGB)** [22]: standardised large-scale graph datasets used for held-out evaluation.

7.2 Evaluation Metrics

Table 2: VDT benchmark metrics.

Metric	**Symbol**	**Measures**
Reconstruction MSE	$\|\mathbf{x} - \hat{\mathbf{x}}\|^2$	Wiring path fidelity
Latent KL	KL_z	Isotropic regularisation
Spectral basis KL	KL_S	Alignment with index $\mathcal{I}$
taumode KL	KL_τ	Frequency band selection
Active modes	N_{active}	Modes with $\mathbb{E}[\omega_k] > 0.01$
Memory SNR	d_k/N_{stored}	Hopfield retrieval quality
ELBO Bayes factor	$\exp(\mathcal{L}(\mathcal{I}_1) - \mathcal{L}(\mathcal{I}_2))$	Index comparison
Linear probe acc.	$\mathrm{Acc}_{\mathrm{probe}}$	Latent discriminability

7.3 Flagship Demo: Entropy-Controlled Spectral Generation

Standard VAEs decode $\mathbf{z}$ into flat feature vectors; the VDT decodes $\mathbf{z}$ into a *graph wiring*—a Laplacian whose eigenvalues are vibrational modes of the spring network (cf. Rayleigh [1]). The latent space directly encodes spectral geometry, enabling samples whose spectral entropy matches a target level. This capability is unique to the VDT's vibrational latent representation and is unavailable in flat-prior models.

8 Related Work

Spectral graph autoencoders. Graph variational autoencoders [15] encode topology in the latent space but do not use spectral priors. VGAE and its successors decode adjacency matrices directly, without a vibrational interpretation.

Probabilistic PCA on graphs. Laplacian Eigenmaps [13] and Diffusion Maps [14] use the heat kernel for dimensionality reduction but are deterministic and lack variational objectives.

NVIB. The Nonparametric Variational Information Bottleneck [18] applies variational compression to sequence models. The VDT shares the ELBO philosophy but applies spectral-structural priors rather than capacity constraints on a fixed-dimension latent.

Beta-VAE and disentanglement. Higgins et al. [12] introduce β-VAE, showing that stronger KL regularisation encourages disentangled latent representations. The VDT's eigenvalue-weighted spectral-basis KL can be seen as an anisotropic generalisation of this pressure, tuning the regularisation strength independently per spectral frequency band.

Real quantum mechanics. Renou et al. [16] and McKague et al. [17] provide experimental and theoretical grounding for real-valued quantum theories. The VDT does not claim to *be* a quantum system, but borrows the structure of signed-amplitude interference as a computational primitive.

9 Discussion and Future Work

The VDT establishes a formal connection between three bodies of theory: Rayleigh's vibrational mechanics, Aaronson's real-amplitude quantum probabilities, and Bayesian graph-spectral learning. The central contribution is the four-term ELBO (10), which replaces hard spectral penalties with proper variational counterparts while preserving the ArrowSpace wiring geometry.

A Spectral Diffusion improvement on this architecture is work in progress.

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```
Input x (B, D)         Frozen: U_q (N, q),  eigvals_q (q,)
    |                              |
    +---[fingerprint from eigvals_q, derived internally]---+
    |
    v
+----------------------------------+
| WiringEncoder                    |
|  MLP + eigenvalue fingerprint    |
|  --> mu, log_var                 |
|      [reparameterise --> z]      |
|  --> log_a, log_b                |
|      [ModeWeightHead --> q(w)]   |
+----------------------------------+
    |
  z_to_q: z (B, latent_dim) --> z_q (B, q)
    |
    v
+------------------------------------------------+
| SpectralLoadingDecoder                         |
|  inputs:  z_q,  U_q,  L_base                  |
|  outputs: W,  omega,  S,  log_var_S,  L_z     |
|  W = U_q @ diag(omega) @ S   (B, N, q)        |
+------------------------------------------------+
         |            |              |
         W       (S, log_var_S)  (log_a, log_b)
         |            |              |
      L_z (B,N,N)   kl_S          kl_tau
         |
  TauModeDiffusion(L_z, E, node_idx)
         |
      x_hat --> recon_loss

loss = recon + kl_z + kl_S + kl_tau
```

Figure 1: VDT v2 data flow (forward signature: `(x, U_q, eigvals_q, node_idx, spectral_cache)`). The index $\mathcal{I}$ enters only through the frozen eigenpair $(\mathbf{U}_q, \boldsymbol{\lambda}_q)$ precomputed once before training; no Laplacian eigendecomposition is performed at training time. The eigenvalue fingerprint is derived *internally* by `WiringEncoder` from `eigvals_q`. `log_var_S` is a separate output of `SpectralLoadingDecoder` required for the KL_S computation (fix #52). Note: N is the graph node count; D is the input feature dimension. These differ in practice (e.g. CORA: N=2708, D=1433).